Change Management for Sustainability

Overcoming the barriers to implementing sustainability initiatives

Learning outcomes

- **Diagnose** why sustainability initiatives stall, using a six-part barrier taxonomy.
- **Apply** Kotter, Lewin and ADKAR to a real sustainability initiative.
- **Quantify** a split-incentive problem, a force-field analysis and an adoption funnel.
- **Design** a change plan: coalition, quick wins, incentives, nudges and KPIs.
- **Evaluate** Interface, Cochin Airport and BP, and the limits of change models.

SECTION 1

Why Initiatives Stall

Most organisations now say sustainability matters. The hard part is turning commitment into daily practice.

The implementation gap

- **Commitment is high:** Deloitte's 2025 global C-suite survey found sustainability embedded in executive priorities, yet impact still lagging.
- **Barriers it names:** limited resources, policy uncertainty, and poor data availability and quality.
- **The core gap:** many firms struggle to turn leadership commitments into day-to-day action.
- **Money is moving:** Deloitte's 2026 survey reports most companies increased sustainability investment, citing higher returns.

SO WHAT?

The bottleneck has shifted from "should we?" to "how do we actually get it done?" That is a change-management problem, not only a technology problem.

The "70% of change fails" myth

70%

The statistic quoted in countless consulting decks and textbooks

5

Published instances Hughes (2011) traced and reviewed

0

Valid, reliable empirical studies he found supporting the figure

Change efforts do fail, but no one knows the true rate. Do not quote 70% as fact; use it only as a prompt to plan carefully.

Why sustainability change is harder than ordinary change

Typical business change

- Benefits land in one department
- Payback is usually visible within a budget cycle
- Owner and metric are clear
- Data exists in existing systems
- Success is judged by profit

Sustainability change

- Benefits spread across functions and years
- Payback may be long or indirect
- Ownership is shared, so often no one's job
- Data (Scope 3, water, waste) is immature
- Success mixes cost, risk, reputation and ethics

A six-part barrier taxonomy

Strategic & leadership

No clear target, weak sponsorship, sustainability seen as a side project.

Structural

Silos, unclear ownership, no cross-functional forum.

Financial

Upfront cost, short payback hurdles, split incentives.

Data & measurement

Poor baselines, inconsistent metrics, weak supplier data.

Cultural & behavioural

Habit, status-quo bias, scepticism, "not my job".

External & policy

Shifting regulation, weak infrastructure, supply chain limits.

Strategic and leadership barriers

- **Vague ambition:** "Be greener" with no baseline, target, owner or budget.
- **Weak sponsorship:** a CEO announcement, but no senior leader whose pay or promotion depends on delivery.
- **Side-project status:** the sustainability team reports far from where capital and operations decisions are made.
- **Short-termism:** quarterly results crowd out multi-year investments.
- **Skills gap:** managers lack the capability to run decarbonisation projects.

DIAGNOSTIC QUESTION

If the sustainability lead left tomorrow, would the initiative continue? If not, it is not yet embedded in the business.

Structural barriers: silos and unclear ownership

- **Cross-cutting by nature:** procurement, facilities, product, finance and IT each hold one piece of the problem.
- **Nobody owns the whole:** each function optimises its own budget and KPI.
- **Handoff failures:** targets set centrally are not translated into plant-level or team-level plans.
- **Slow decisions:** approvals pass through many committees, so pilots die in the queue.
- **Fix direction:** cross-functional steering group, named owners, targets cascaded into each unit.

Financial barriers: cost, payback and split incentives

- **Upfront capital:** efficiency projects compete with growth projects that promise higher returns.
- **Hurdle rates:** firms demand payback of 2-3 years even when 10-year savings are large.
- **Split incentives:** the unit that pays for the upgrade is not the one that enjoys the lower bill.
- **Hidden benefits:** risk reduction, talent retention and brand value rarely appear in the spreadsheet.
- **Fix direction:** life-cycle costing, internal carbon price, shared-savings or green-fund mechanisms.

ENERGY-EFFICIENCY GAP

Many efficiency measures with attractive returns are still not adopted. Cost is only part of the story; incentives and attention matter too.

Worked example: a split incentive on campus

PROBLEM A campus replaces 2,000 tube lights (36 W) with LEDs (18 W). Use 3,000 h/yr, tariff Rs 8/kWh, cost Rs 450 per fitting (illustrative figures), 5 years, 12% discount rate.

1. Power saved = 2,000 x (36 - 18) W = 36,000 W = 36 kW
2. Energy saved = 36 kW x 3,000 h = 108,000 kWh per year
3. Annual saving = 108,000 x Rs 8 = Rs 8.64 lakh
4. Capital cost = 2,000 x Rs 450 = Rs 9.0 lakh; simple payback = 9.0 / 8.64 = 1.04 years
5. PV of 5 years of savings = 8.64 x 3.605 (annuity factor) = Rs 31.1 lakh; NPV = 31.1 - 9.0 = Rs 22.1 lakh
6. The barrier: if the department budget pays Rs 9 lakh but the central estate pays the bill, the department sees only cost.

Strong economics, yet it stalls. The fix is organisational: pool budgets or share savings.

Data and measurement barriers

- **No baseline:** you cannot show progress without a starting point.
- **Inconsistent metrics:** different units and boundaries make results incomparable across sites.
- **Scope 3 opacity:** supplier and customer emissions are estimated, not metered.
- **Manual collection:** spreadsheets and email chains breed errors and delay.
- **Consequence:** leaders do not trust the numbers, so they do not fund or act on them.
- **Fix direction:** start with a few reliable metrics, automate data capture, assign data owners.

Cultural and behavioural barriers

Status-quo bias

People prefer the familiar. "We have always done it this way."

Present bias

Costs are now, benefits later, so action keeps getting postponed.

Loss aversion

Giving up a convenience feels larger than an equal gain.

"Not my job"

Responsibility diffuses when everyone is partly responsible.

Cynicism

Past greenwashing or failed programmes make staff discount new ones.

Change fatigue

Too many parallel initiatives exhaust attention and goodwill.

External and policy barriers

- **Regulatory uncertainty:** fragmented or shifting rules make long investments hard to plan.
- **Infrastructure limits:** weak grid, no recycling system or no charging network outside the firm's control.
- **Supplier readiness:** small suppliers lack data, capital or skills to follow your targets.
- **Market signals:** customers say they want green products but may not pay a premium.
- **Response:** scenario planning, industry coalitions, supplier support programmes, policy engagement.

Activity: sort the barrier

Place each statement into the barrier family that best fits. Discuss in pairs for 3 minutes.

Strategic / Structural	Financial / Data	Cultural / External
Targets, sponsorship, ownership, silos, decision paths	Cost, payback, incentives, baselines, metrics	Habits, trust, regulation, infrastructure, suppliers

1. "The CEO announced net zero but nobody's bonus depends on it."
2. "Facilities pays for the chillers; the departments pay the power bill."
3. "Our site data is in five different spreadsheet formats."
4. "Staff roll their eyes; last year's green drive just vanished."
5. "The state recycling facility is 400 km away."
6. "Procurement and design never talk to each other."

Quick check 1

According to Hughes (2011), what is the evidence for the claim that 70% of organisational change initiatives fail?

A	A. Strong: it comes from many large randomised studies
B	B. Mixed: some studies support it, some contradict it
C	C. None valid or reliable: the figure lacks empirical support
D	D. It was proven for sustainability initiatives only

Think, vote, then discuss. (Answer in speaker notes.)

Quick check 1: answer

Answer: C. No valid, reliable empirical support

Hughes (2011) examined five published uses of the 70% figure and found no empirical basis. Lesson: be sceptical of precise-sounding statistics. Failure of change efforts is common and real, but its true rate is unknown, so plan for risk without quoting a made-up number.

SECTION 2

Frameworks for Leading Change

Three classic models give you a vocabulary for diagnosing a stalled initiative and choosing the next move.

What is change management?

- **Definition:** the structured approach to moving people, teams and organisations from a current state to a desired future state.
- **Technical change:** install the solar panels, change the process. The solution is known.
- **Adaptive change:** people must alter habits, values and relationships. No quick fix exists.
- **Sustainability is mostly adaptive:** the technology often exists; behaviour, incentives and power structures are the obstacle.
- **Two sides:** the project side (scope, schedule, budget) and the people side (adoption, trust, habits).

Kotter's 8 steps

1. Create urgency

Make the case for acting now.

2. Build a coalition

Assemble a guiding team with influence.

3. Form a vision

A clear picture of the future and strategy.

4. Communicate

Enlist volunteers with a simple, repeated message.

5. Remove obstacles

Clear barriers; empower action.

6. Quick wins

Plan and celebrate visible short-term wins.

7. Build on gains

Do not declare victory too early.

8. Anchor in culture

Embed new practices in norms and systems.

Kotter applied to a campus net-zero programme

Step	What it looks like in practice
Urgency	Share the campus energy bill, emissions trend and regulatory outlook with leadership
Coalition	Dean, estates head, finance officer, student leaders, two respected faculty
Vision	"A campus that cuts its electricity emissions by half in five years while saving money"
Communicate	Monthly dashboards, hostel competitions, town halls, simple slogans
Obstacles	Pool the energy budget, simplify procurement for efficiency items
Quick wins	LED retrofit in two blocks, visible savings in the first quarter
Build on gains	Extend to HVAC, rooftop solar and hostels
Anchor	Green procurement policy, energy KPIs in building managers' reviews

Lewin: unfreeze, change, refreeze

Unfreeze

Create readiness. Show why the current state is unsustainable and lower anxiety about the move.

Change

Move to new behaviours and processes, with training, support and experimentation.

Refreeze

Stabilise the new state with policies, incentives, routines and feedback.

Critique: real organisations rarely sit still, so treat refreeze as "make the new way the default" rather than "stop changing".

Worked example: force-field analysis (Lewin)

PROBLEM Should a firm launch a plant-wide energy-management system? Score each force from 1 (weak) to 5 (strong).

Driving forces: regulation 4, cost savings 4, customer demand 3, employee pride 2 = 13

Restraining forces: upfront cost 4, no reliable data 3, time on shop floor 3, scepticism 3, silos 2 = 15

Net = 13 - 15 = -2, so the change is likely to stall as designed.

Strategy: reducing restraining forces is usually cheaper than pushing harder on driving forces.

Action: run a 3-month pilot (scepticism 3 to 1) and install sub-meters (data 3 to 1).

New restraining total = 4 + 1 + 3 + 1 + 2 = 11, so net = 13 - 11 = +2.

Same drivers, but the change moves from -2 to +2 by addressing resistance, not pushing harder.

Prosci ADKAR: change at the individual level

Awareness

Of the need for change

Desire

To support and take part

Knowledge

Of how to change

Ability

To implement skills and behaviours

Reinforcement

To sustain the change

Using ADKAR as a diagnostic

Symptom you observe	Likely gap	Intervention
"Why are we doing this?"	Awareness	Share data and the business case; explain the why repeatedly
"I get it, but it is not my problem."	Desire	Link to personal goals, involve staff, address what they lose
"I do not know how to do this."	Knowledge	Training, guides, worked examples
"I tried but the tools do not work."	Ability	Coaching, time to practise, fix process and tools
"It lasted a month, then people drifted back."	Reinforcement	Recognition, KPIs, defaults, manager follow-up

Resistance: enemy or information?

Resistance as enemy

- Overcome it with a stronger push
- Label sceptics as "blockers"
- Communicate only benefits
- Result: compliance without commitment

Resistance as information

- Ask what the person is protecting
- Often points to real design flaws
- Involve critics in solving the issue
- Result: a better plan and real buy-in

Stakeholder map: power and interest

Group	Power / interest	Engagement approach
Senior leadership, CFO	High / varies	Manage closely: business case, risk framing, KPIs
Facility and plant managers	High / high	Co-design the plan; they carry the delivery
Finance and procurement	Medium / low	Show how rules (hurdle rates, tenders) block the goal
Employees, students	Low / medium	Inform, involve, give feedback and recognition
Suppliers, regulators, community	Varies	Collaborate, monitor, support capability

Comparing the three models

Model	Level	Strength	Limitation
Kotter 8 steps	Organisation	Practical sequence for large programmes	Linear; little on individual adoption
Lewin	Organisation / group	Simple; force-field analysis is a quick diagnostic	Static view of a fluid world
ADKAR	Individual	Pinpoints where a person or team is stuck	Less on strategy and power politics

Use them together: Kotter to plan, Lewin to diagnose forces, ADKAR to check people.

Quick check 2

Staff understand why the new waste-sorting scheme matters and know how to do it, but still do not follow it. Which ADKAR element is most likely missing?

A	A. Awareness
B	B. Desire
C	C. Knowledge
D	D. Reinforcement only

Think, vote, then discuss. (Answer in speaker notes.)

Quick check 2: answer

Answer: B. Desire

They are aware and they know how, so the problem is motivation. Respond by asking what they lose or fear, linking the scheme to their own goals, involving them in design, and fixing any convenience problems. More training or posters would not help here.

SECTION 3

Leading Change in Practice

Plans do not fail on paper. They fail in meeting rooms, budgets and daily habits. This section is about the practical moves.

Build the case and tell the story

- **Speak each audience's language:** cost and risk for finance, brand and growth for marketing, uptime for operations.
- **Pair numbers with a narrative:** a clear "from-to" story people can repeat.
- **Be honest about trade-offs:** name what will get harder and what will be given up.
- **Use life-cycle costs:** show total cost of ownership, not only purchase price.
- **Make the first step small:** a specific, easy ask beats a grand vision.

TEST YOUR PITCH

Can a mid-level manager explain, in two sentences, why the initiative matters and what is expected of them?

Coalition, sponsors and champions



Executive sponsor

Owns the target, unblocks budgets and signals priority.



Steering group

Cross-functional leaders who resolve conflicts between units.



Champion network

Respected people in each team who adapt and promote the change locally.



Feedback loops

Surveys, drop-ins and forums that surface problems early.

Sequencing: quick wins first, then scale



A quick win must be visible, unambiguous and linked to the goal, otherwise it builds cynicism, not momentum.

Incentives: what gets rewarded gets done

Misaligned

- Managers rewarded only for short-term cost cuts
- Sustainability team measured by reports, not outcomes
- Procurement paid on lowest price
- Result: green goals lose every trade-off

Aligned

- A share of bonus linked to emissions or efficiency targets
- Cost of carbon in investment appraisal
- Procurement scored on total cost and supplier data
- Result: the green choice becomes the rational one

Nudges: make the right thing easy



Defaults

Double-sided printing, green electricity or vegetarian option set as the default.



Feedback

Show real-time energy use and comparison with similar buildings.



Social norms

Tell people what most colleagues already do.



Reduce friction

Put recycling bins where the waste is created; automate what you can.

Worked example: why the weakest link matters (ADKAR funnel)

PROBLEM A firm launches a "switch off at the end of day" programme for 1,000 employees. Share who clear each ADKAR stage: A 90%, D 60%, K 80%, A 75%, R 70%.

1. Sustained adoption = 0.90 x 0.60 x 0.80 x 0.75 x 0.70 = 0.2268, about 227 people (22.7%).
2. Option 1: boost Awareness from 90% to 100% (more emails): 1.00 x 0.60 x 0.80 x 0.75 x 0.70 = 25.2%.
3. Option 2: boost Desire from 60% to 80% (involve staff, address concerns): 0.90 x 0.80 x 0.80 x 0.75 x 0.70 = 30.2%.
4. Same effort, but fixing the weakest stage gives +7.5 points versus +2.5 points for the strongest stage.
5. Caveat: stages are not truly independent; this is a model for prioritising, not a forecast.

Spend effort where the funnel narrows most, not where it is easiest to act.

Case: Interface and Mission Zero

LEADERSHIP CONVICTION OVER 25 YEARS

- 1994: founder Ray Anderson has his "spear in the chest" moment and commits the carpet maker to zero negative impact by 2020.
- Mission Zero set aggressive targets across factories, products and supply chain.
- Nov 2019: company announced success ahead of schedule.
- Reported results: carpet carbon footprint down 74% since 1996; market-based GHG intensity at carpet factories down 96%.
- Successor goal: Climate Take Back, aiming to reverse global warming.

LESSON FOR CHANGE LEADERS

A credible sponsor, a public and ambitious target, and persistence across leadership changes. Check how "zero" is defined before treating it as a model for your own claims.

Case: Cochin International Airport goes solar

START SMALL, THEN SCALE

- Started with a small 1 MW solar plant, then learned from it.
- Aug 2015: commissioned a 12 MW solar PV plant, 46,150 panels over 18 hectares, cost about Rs 65 crore.
- Expected output about 18 million units a year; more than 300,000 t CO2 avoided over 25 years.
- Built with an engineering partner under the airport company, backed by state government.
- Caveat: "fully solar" is an annual balance. The airport still uses the grid when the sun is down.

LESSON FOR CHANGE LEADERS

A small pilot reduced risk and built confidence. A clear economic story (the power bill) and committed leadership carried the scale-up. Be exact about claims.

Case: BP, when commitment retreats

HONEST CRITIQUE

- 2020: BP announced a net-zero ambition and a plan to cut oil and gas output by 2030 while expanding low-carbon investment.
- 2023: the oil and gas reduction target was scaled back.
- 2025: BP reset its strategy, cutting low-carbon spending and raising oil and gas investment.
- Pressures cited: shareholder returns, share price, energy security and changing policy.
- Verify the exact figures from BP's own statements before quoting.

LESSON FOR CHANGE LEADERS

Commitments that are not anchored in the core business model, incentives and investor expectations can reverse. Sustainability change is exposed to market and political cycles.

Discussion: a change plan for IIT Delhi

Scenario: the Institute wants to cut hostel and lab electricity use by 30% in three years. Departments pay their own bills, students see no cost, and no one owns the target.

1. Which barrier families are present? Rank them.
2. Who are the key stakeholders and what does each lose?
3. Which quick win would you run in the first 90 days?
4. Which incentive or default would you change?

SECTION 4

Making Change Stick

A successful pilot is not a successful change. The test is whether it survives budget cuts, leadership turnover and the next shiny priority.

Anchoring change in the organisation

- **Policy and procedure:** revise procurement rules, travel policy, design standards.
- **Budgets:** a dedicated, protected funding line or internal carbon price.
- **People systems:** sustainability duties in job descriptions, appraisals and onboarding.
- **Governance:** regular board-level review with public targets.
- **Culture:** stories, recognition and leaders who visibly follow the new norms.

REFREEZE

Embedding means the new way becomes "how we do things here" rather than a campaign needing constant energy.

Measuring change: leading and lagging indicators

Type	Examples	Use
Leading (activity)	Training completion, pilots launched, suppliers with data, ideas submitted	Early warning on whether the change is taking hold
Adoption	Share of units reporting data, % procurement with green criteria	Shows behaviour actually changing
Lagging (outcome)	kWh per unit, tCO2e, waste diverted, cost saved	Confirms real impact and supports the business case
Sentiment	Pulse surveys on trust, workload, change fatigue	Spots resistance and fatigue before they harden

Backsliding and fatigue vs. durable change

Warning signs

- Declaring victory after the pilot
- Champions overloaded, no time or recognition
- Targets met by reporting changes or offsets only
- Messages repeat but nothing visibly changes

Countermeasures

- Fix reinforcement: KPIs, defaults, manager follow-up
- Reduce competing initiatives; sequence them
- Audit data and claims to protect trust
- Share honest progress, including setbacks

Critique: the limits of change-management models

- **Evidence is thin:** frameworks are popular but rigorous comparative evidence is limited, as the 70% myth shows.
- **Power is under-weighted:** models treat resistance as a communication problem; sometimes interests really conflict.
- **Context matters:** a model tested in a corporation may not suit a university, a government or a village cooperative.
- **Risk of manipulation:** managing people's "buy-in" can mask a poor decision; sometimes the initiative should change, not the people.
- **Systemic limits:** one firm cannot fix prices, policy or infrastructure; collective action is also needed.

Building a Sustainability Culture

Engaging employees and creating a shared vision for sustainability

Learning outcomes

- **Explain** what a sustainability culture is, using Schein's three levels.
- **Craft** a shared vision that is clear, credible, compelling and connected.
- **Diagnose** employee behaviour with motivation theory and the COM-B model.
- **Design** engagement mechanisms: communication, participation, recognition, defaults.
- **Measure** culture with a pulse survey, and judge cases and limits critically.

Roadmap for today



1. Culture and vision

What culture is, and how to build a vision people share.



2. Engaging employees

Motivation, behaviour, communication, leaders, two cases.



3. Embedding in practice

Co-creation, rituals, rewards, measurement.



4. Sustaining the culture

Authenticity, fatigue, subcultures, critique.

SECTION 1

Culture and Shared Vision

Policies tell people what to do. Culture decides what they do when no one is watching.

What is a sustainability culture?

- **Culture:** the shared values, beliefs, norms and habits that shape "how things are done here".
- **Sustainability culture:** one where environmental and social impact is considered in everyday decisions, by everyone, without being told.
- **Not the same as:** a sustainability department, a policy document or an annual report.
- **Why it matters:** thousands of small decisions (what to buy, print, switch off, design) are made far from any manager.
- **Test:** what happens when sustainability conflicts with a deadline or a cost target?

KEY IDEA

You cannot mandate culture. You can shape it through leadership behaviour, systems and shared meaning.

Schein's three levels of culture



Artifacts

Visible: recycling bins, dashboards, dress, rituals, reports. Easy to see, hard to interpret.



Espoused values

Stated: mission, values, "we care about the planet". What the company says.



Basic assumptions

Unspoken beliefs taken for granted: "growth comes first", "cheap is smart". They drive real behaviour.

The say-do gap: stated values vs. lived assumptions

Espoused value	Observed behaviour	Underlying assumption
"We care about the planet"	Procurement picks the lowest bid every time	Cost is the only real measure
"Everyone is a sustainability ambassador"	No time or budget given for it	Sustainability is extra work
"Net zero by 2040"	Bonuses depend only on quarterly profit	Short-term results come first
"Innovation for good"	Failed green pilots are blamed on individuals	Risk-taking is unsafe

Staff believe the assumptions, not the posters. Change the assumption by changing what is rewarded and tolerated.

Culture types and what each means for sustainability

Culture type (Competing Values)	Typical strength	Typical sustainability risk
Clan (collaborative)	Strong buy-in, shared values, volunteering	Groupthink; avoids hard trade-offs
Adhocracy (innovative)	Quick pilots, new products and models	Pilots without scale or follow-through
Market (results-driven)	Clear targets, accountability	Only what is measured and monetised counts
Hierarchy (rule-based)	Reliable compliance and procedures	Slow change; compliance mindset

No type is best. Know your starting culture and build on its strengths.

A maturity ladder for sustainability culture



Compliance

Meet the rules. Sustainability is a legal duty handled by a few specialists.

Programmes

Campaigns and projects run by a team. Enthusiasm depends on a few people.

Embedded

Built into targets, processes and job roles. Staff raise ideas unprompted.

Identity

Part of who we are. Shapes strategy, hiring and partnerships.

A teaching model, not a measured standard. Most organisations sit at different levels in different units.

Why a shared vision matters

- **Compliance vs. commitment:** Senge argues a genuinely shared vision creates commitment, whereas imposed goals get only compliance.
- **Alignment:** a common picture lets teams make consistent decisions without constant instruction.
- **Energy:** people work harder for a goal they helped shape and understand.
- **Direction in conflict:** when two good options clash, the vision breaks the tie.
- **Risk of the opposite:** a slogan written by a consultancy and ignored by staff breeds cynicism.

SHARED, NOT ANNOUNCED

A vision is shared when staff can describe it in their own words and link it to their own work.

Anatomy of a good vision: the four Cs



Clear

Simple enough to repeat. Specific about where and by when.



Credible

Backed by a baseline, a plan and leaders' own actions.



Compelling

Connects to values people care about, not only cost.



Connected

Each team can see its own part in the whole.

Worked example: from vision to measurable milestones

PROBLEM Vision: "Halve the campus's electricity emissions in five years." Baseline 15 GWh/yr, grid factor 0.71 tCO2/MWh, solar yield 1.4 GWh per MWp (illustrative assumptions).

1. Baseline emissions = 15,000 MWh x 0.71 = 10,650 tCO2 per year
2. Target = 50% cut, so 7.5 GWh avoided (holding the grid factor constant); emissions avoided = 7.5 GWh x 0.71 = 5,325 tCO2
3. Efficiency (LED, controls, behaviour) at 20% of 15 GWh = 3.0 GWh
4. Remaining 7.5 - 3.0 = 4.5 GWh from solar; capacity = 4.5 / 1.4 = 3.2 MWp
5. Milestone: 7.5 / 5 = 1.5 GWh avoided per year, on average, tracked monthly
6. Translate for each block or hostel: share of the 1.5 GWh and a named owner, so the vision becomes a local goal.

A vision becomes shared when each unit can see its own number, and one person owns it.

Quick check 1

Staff say, "Management talks green, but we all know the cheapest quote always wins." At which of Schein's levels does the real problem sit?

A	A. Artifacts
B	B. Espoused values
C	C. Basic assumptions
D	D. None; culture is not layered

Think, vote, then discuss. (Answer in speaker notes.)

Quick check 1: answer

Answer: C. Basic assumptions

The espoused value (green) is visible in statements, but the underlying assumption (cheapest wins) drives actual procurement. Fix it by changing the evaluation criteria (total cost, supplier data) and by backing a manager who chooses a greener, costlier bid.

Activity: which level of culture?

Place each observation into one of Schein's levels. Discuss in pairs for 3 minutes.

Artifact	Espoused value	Basic assumption
Visible objects, rituals, dashboards, spaces	Stated principles, mission, slogans	Taken-for-granted, unspoken beliefs

1. "Energy dashboard displayed in the lobby."
2. "Our mission: leave the planet better than we found it."
3. "Of course the plant runs flat out; savings projects can wait."
4. "Monthly zero-waste day with a team prize."
5. "We value people, planet and profit equally" (annual report).
6. "Nobody questions a senior person who ignores the rules."

SECTION 2

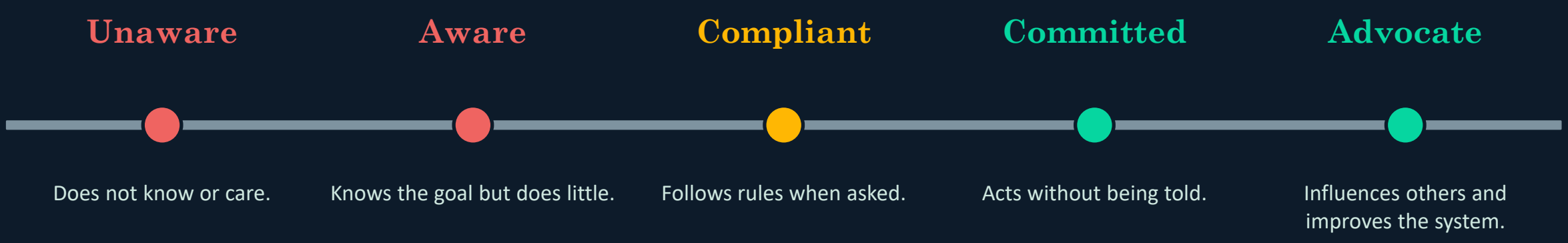
Engaging Employees

Employees are not an audience for sustainability communications. They are the people who carry it out.

Why employees are central

- **Scale of decisions:** staff make most operational choices: energy use, materials, waste, design details.
- **Frontline knowledge:** people closest to the process see the waste and the quick fixes.
- **Delivery:** targets set at the top are met or missed in teams.
- **Talent:** surveys, including Deloitte work on retail workers, link employer sustainability efforts with retention and attraction; check findings for your sector.
- **Credibility:** employees are also the first to spot greenwashing.

The engagement spectrum



Moving people one step is realistic. A campaign that expects everyone to become an advocate will disappoint.

What motivates people: self-determination and purpose



Autonomy

Choice over how to contribute. Mandates without choice breed resentment.



Competence

Feeling able and skilled: training, tools, visible progress.



Relatedness

Belonging to a team that cares: green teams, shared goals.



Purpose

Seeing how my work links to something bigger.

The COM-B model: why do people not act?

Component	Question	Typical sustainability fix
Capability	Do they have the knowledge and skills?	Training, simple guides, clear labelling
Opportunity	Do the environment and systems make it possible?	Bins, bike racks, sub-meters, default settings, time
Motivation	Do they want to, and do they believe it matters?	Purpose, feedback, recognition, peer norms

Source: Michie, van Stralen and West (2011). Diagnose all three before choosing the intervention.

Communication: broadcast or dialogue?

Broadcast only

- Posters, emails, annual reports
- One-way, same message for all
- Measures reach, not understanding
- Result: awareness without change

Dialogue

- Town halls, team huddles, Q&A, listening sessions
- Messages adapted to each role
- Asks what blocks people and acts on it
- Result: trust and ownership

Storytelling and framing

- **Local over global:** link climate to the team's own site, product or campus.
- **Stories over statistics:** one colleague's idea and result is remembered longer than a chart.
- **Show the ask:** say exactly what each person should do next.
- **Frame to values:** cost and efficiency for some, health, fairness or pride for others.
- **Avoid:** guilt and doom as the main message; they trigger avoidance.

REPEAT, VARY, SHOW

People need to hear a message many times, through different channels, and see leaders act on it.

Ways to involve employees



Green teams

Volunteer cross-functional groups that run local projects.



Ideas platform

A simple channel to submit, vote and track ideas.



Hackathons

Time-boxed events to solve a real sustainability problem.



Town halls

Open Q&A with leaders about progress and difficulties.



Ambassadors

Trained peers who explain and encourage in their teams.



Volunteering

Community and restoration days that build pride and bonds.

Worked example: campaign vs. default (shut-down behaviour)

PROBLEM 500 office PCs left on overnight draw 60 W for 14 idle hours, 250 working days a year. Tariff Rs 8/kWh, grid factor 0.71 kgCO2/kWh (illustrative).

1. Waste per PC = 0.060 kW x 14 h x 250 days = 210 kWh per year
2. Fleet waste = 210 x 500 = 105,000 kWh per year
3. Campaign: 60% of staff comply, so saving = 63,000 kWh = Rs 5.04 lakh and 63,000 x 0.71 = 44.7 tCO2
4. Default: auto sleep/shut-down at 95% effectiveness saves 99,750 kWh = Rs 7.98 lakh and 70.8 tCO2
5. The default saves about 58% more than the campaign (99,750 / 63,000 = 1.58), with no ongoing effort.
6. Use engagement for what defaults cannot do: ideas, ownership, and acceptance of the default.

Engagement and system design work together: set the default, then explain why.

Case: Unilever Sustainable Living Plan

A COMPANY-WIDE VISION

- 2010: launched under CEO Paul Polman with goals for health and wellbeing, a smaller environmental footprint and better livelihoods.
- Positioned as a plan for growth, tying the vision to brands and to employees' sense of purpose.
- Replaced in 2020 by the Unilever Compass, a broader strategy.
- Later reports say some targets were softened or dropped under new leadership. Verify the details.

LESSON FOR CHANGE LEADERS

A bold, public vision can rally employees and attract talent. But if the vision is tied to one leader and not to systems and incentives, it can fade when leadership changes.

Case: ITC, an Indian triple-bottom-line culture

TARGETS EMBEDDED IN THE BUSINESS

- ITC reports being carbon positive, water positive and solid-waste-recycling positive for many consecutive years.
- The goals are set at enterprise level and applied across diverse businesses.
- Sustainability is presented as part of how the company creates value.
- Honest critique: the core portfolio includes tobacco, so critics question how far such leadership offsets that impact.
- Check the latest ITC report for current years and figures.

LESSON FOR CHANGE LEADERS

Concrete, measurable "positive" goals are easy for employees to grasp. A credible culture also needs honesty about the hardest trade-offs in the portfolio.

Leaders set the culture



Say

Explain why sustainability matters here, in plain words, repeatedly.



Do

Visible behaviour: travel choices, meeting habits, how trade-offs are made.



Fund

Give time, budget and authority to the people doing the work.



Listen

Seek bad news and criticism; act on it publicly.

Middle managers: the squeezed middle

- **Position:** they translate strategy into daily work and carry the targets that conflict.
- **Common problem:** told to "be sustainable" but judged only on output and cost.
- **Result:** quiet deprioritisation, even when they agree with the goal.
- **What helps:** clear targets in their own scorecard, training and a say in how to meet them.
- **Why it matters:** staff copy what their immediate manager does and rewards.

Quick check 2

Employees at a firm want to cycle to work and know the benefits, but there is no secure bike parking or shower. Which COM-B component is the main gap?

A	A. Capability
B	B. Opportunity
C	C. Motivation
D	D. None; it is a values problem

Think, vote, then discuss. (Answer in speaker notes.)

Quick check 2: answer

Answer: B. Opportunity

Knowledge (capability) and willingness (motivation) are already present. The missing piece is physical: secure parking, showers, safe routes. More posters will not help. Provide the facilities, then promote them.

SECTION 3

Embedding in Practice

Moving from words to a way of working: co-creation, rituals, rewards and measurement.

Co-creating the vision



Listen

Interviews, surveys and workshops across levels and sites.

Draft

A small cross-functional team writes options in plain language.

Test

Share drafts, ask "does this feel true? what is missing?"

Launch

Leaders commit, with first-year milestones and owners.

The process matters as much as the words. A vision people helped write is one they defend.

Vision workshop: a half-day format

Activity	Time	Output
Current-state reality check: where do we stand, honestly?	30 min	Shared fact base and baseline
Future-state story: "It is 2032, what does our campus or site look like?"	45 min	Draft images of success
Cluster themes and test against the four Cs	45 min	3-5 vision themes
Barriers and enablers (force-field)	45 min	Top obstacles and supports
Commitments: what will each team do in 90 days?	45 min	Owners and first actions

Rituals, symbols and language

- **Rituals:** a short sustainability item at every team meeting; an annual "green day" with real projects.
- **Symbols:** who gets the reserved parking, the awards, the stage time; reuse and repair made visible.
- **Language:** swap "cost of green" for "cost of waste"; make "ask me about carbon" a normal sentence.
- **Space:** dashboards in common areas, plants, bike racks near the front door.
- **Warning:** artifacts without behaviour behind them become jokes.

Systems that signal what really matters

Systems that undermine

- Hiring and induction never mention sustainability
- Promotions ignore it
- Travel and procurement rules reward the cheapest only
- Budgets reset yearly, killing multi-year projects

Systems that reinforce

- Induction includes the vision and each person's role
- Sustainability is in job descriptions and appraisals
- Procurement scores total cost and supplier data
- Protected multi-year funding line

Recognition and rewards



Celebrate teams

Prize the team, share its method, not just the winner.



Link to appraisal

Include contributions in performance reviews and promotions.



Reinvest savings

Return part of savings to the team for its next idea.



Public thanks

Leaders name individuals and describe what they changed.

Inclusive engagement: reach everyone

- **Frontline and shift workers:** often have no email or meeting time; use toolbox talks and supervisors.
- **Contract and outsourced staff:** cleaners, security, canteen staff touch waste and energy daily; include them.
- **Language and literacy:** use local languages, pictures and demonstrations.
- **Remote and multi-site teams:** do not centre everything at headquarters.
- **Equity link:** a culture that excludes workers from decisions contradicts the social side of sustainability.

Measuring culture: what to track

Dimension	Example indicator	Source
Awareness	% who can state the vision and their own role	Pulse survey
Participation	Green team members, ideas submitted and implemented	Platform data
Leadership credibility	"Leaders act on what they say" (1-5)	Pulse survey
Behaviour	kWh per head, waste diverted, print volumes	Meters, logs
Resources	"I have the time and tools to act" (1-5)	Pulse survey

Worked example: scoring a culture pulse survey

PROBLEM Five items on a 1-5 scale (mean scores): Awareness 4.2, Personal relevance 3.6, Leadership credibility 2.9, Ability and resources 3.1, Peer norms 3.8.

1. Mean score = (4.2 + 3.6 + 2.9 + 3.1 + 3.8) / 5 = 17.6 / 5 = 3.52
2. Convert to a 0-100 index = (3.52 - 1) / 4 x 100 = 63
3. Lowest item: Leadership credibility (2.9). Next lowest: Ability and resources (3.1).
4. Insight: people know and care, but doubt leaders and lack means. More awareness posters would not help.
5. Action: leaders publish and act on one visible commitment; free up time and tools for teams.
6. If credibility rises to 3.9, mean = 18.6 / 5 = 3.72, index = (3.72 - 1) / 4 x 100 = 68.

Read the gaps, not the average: the weakest dimension tells you where to act.

Discussion: a shared vision for IIT Delhi hostels

Scenario: hostel electricity and food waste are rising. Students see no bill, wardens report to different offices, and an earlier "green hostel" poster campaign faded in two months.

1. Which level of culture (Schein) is the real problem?
2. Use COM-B: what is missing for students and for staff?
3. Who should co-create the vision, and how?
4. Which default, ritual or reward would you introduce first?

SECTION 4

Sustaining the Culture

Culture is fragile. It is built slowly and can be damaged quickly by one example of hypocrisy.

Authenticity vs. greenwashing inside the firm

Signs of internal greenwashing

- Slogans without budget or owner
- Targets that quietly move
- Success stories hide failures
- Staff are asked to act while leaders fly business class

Signs of authenticity

- Honest reporting of misses
- Leaders accept trade-offs publicly
- Resources match ambition
- Staff criticism is welcomed and acted on

Change fatigue and champion burnout

- **Too many initiatives:** every campaign competes for the same attention.
- **Volunteer overload:** green team members do this on top of a full job.
- **Recognition gap:** no credit, no time, no budget leads to quiet exit.
- **Remedies:** sequence campaigns, count green-team work as part of the role, rotate leadership, celebrate wins.
- **Watch:** participation falling over time is an early warning of fatigue.

Subcultures, hierarchy and the Indian context

- **Subcultures:** engineering, finance and operations may each interpret sustainability differently; speak their languages.
- **Hierarchy:** where authority is respected, leaders' visible action carries extra weight, and staff may not voice dissent.
- **Frugality as an asset:** traditions of reuse and thrift (often called jugaad or "make do") can be tapped, but must not be confused with real innovation.
- **Diversity:** a single national or global campaign should be adapted by region and site.
- **Caution:** these are tendencies, not rules about any person.

Critique: the limits of culture-building

- **Culture is hard to measure:** surveys capture attitudes, and attitudes are not behaviour.
- **Individual responsibility trap:** asking staff to switch off lights can distract from large structural emissions (energy source, products, supply chain).
- **Weak causal evidence:** many engagement claims rest on case studies and self-reports.
- **Manipulation risk:** engineered "buy-in" may hide poor decisions or real conflicts of interest.
- **Limits of the firm:** culture cannot replace regulation, pricing and infrastructure.

Leadership and Sustainability

Developing leadership skills for driving sustainability within the organisation

Learning outcomes

- **Define** sustainability leadership and distinguish technical from adaptive challenges.
- **Compare** transformational, servant, adaptive, ethical and systems leadership.
- **Apply** core skills: strategic thinking, influence, negotiation and ethical judgment.
- **Quantify** an internal carbon price decision and a 360-degree leadership gap.
- **Plan** your own leadership development, and critique the "heroic leader" view.

SECTION 1

Foundations of Leadership

Technology and targets exist. What is scarce is people who can mobilise others to deliver them.

What is sustainability leadership?

- **Definition:** influencing and enabling others to pursue economic, social and environmental goals together, over the long term.
- **Not just seniority:** leadership is an activity that anyone can exercise, from a CEO to a final-year student.
- **Distinctive features:** long time horizons, many stakeholders, uncertainty and real value conflicts.
- **Research view:** Visser and Courtice (2011) review how leadership theories apply to sustainability.

KEY IDEA

Sustainability leaders do not only manage projects. They help organisations decide what matters and mobilise people to act on it.

Leadership styles and what each adds

Style	What it contributes	Watch out for
Transformational	Inspires a shared vision beyond self-interest	Charisma without substance; dependence on one person
Servant	Puts people and community needs first	Slow decisions; may avoid hard calls
Adaptive (Heifetz)	Mobilises people to face tough change in values and habits	Can unsettle people; needs trust
Ethical / authentic	Aligns actions with values; builds credibility	Moral certainty that shuts out dissent
Systems	Sees interconnections; convenes across boundaries	Complexity paralysis; hard to show quick wins

Effective leaders borrow from all of these depending on the situation.

Technical vs. adaptive challenges (Heifetz)

Technical problem

- Clear definition, known solution
- Experts can solve it
- Example: install LED lighting or rooftop solar
- Authority and a project plan are enough

Adaptive challenge

- Problem and solution both unclear
- People must change habits, loyalties or beliefs
- Example: departments sharing budgets for a joint goal
- Needs learning, dialogue and experiment

Systems leadership: three core capabilities



See the larger system

Help people from different organisations understand the whole, beyond their own part.



Foster reflection

Create conversations where people listen, reflect and rethink their assumptions.



Co-create the future

Shift from reacting to problems to building shared options for the future.

Key competencies for sustainability (Wiek et al.)

Systems thinking

Analyse links across scales and sectors.

Anticipatory

Envision and assess possible futures.

Normative

Judge goals and trade-offs against values.

Strategic

Design and implement change that works.

Interpersonal

Collaborate, negotiate and build trust.

The inner side of leadership



Values

Clarity about what you will not trade away.



Long-term thinking

Willingness to accept costs now for benefits you may not see.



Humility

Admit what you do not know; learn from those with less power.



Moral courage

Speak up and act when it is uncomfortable or costly.

Four levels of leadership

Self

Understand your values, strengths and blind spots; model the behaviour.

Team

Build trust, set goals, create space for ideas and dissent.

Organisation

Shape strategy, structures, incentives and culture.

Ecosystem

Work with suppliers, regulators, peers and communities.

Sustainability challenges rarely respect organisational boundaries, so leaders must work at all four levels.

Quick check 1

A university wants every department to share one energy budget so that savings can be reinvested. Departments refuse, fearing loss of control. This is mainly a:

A	A. Technical problem needing a better meter
B	B. Adaptive challenge needing dialogue and changed assumptions
C	C. Compliance problem needing a stricter rule
D	D. Data problem needing more surveys

Think, vote, then discuss. (Answer in speaker notes.)

Quick check 1: answer

Answer: B. An adaptive challenge

The technology and the arithmetic are known. The obstacle is how people feel about losing control and trusting a shared pool. A leader must convene departments, surface fears, test small shared-budget pilots and build trust. Forcing the rule alone may bring compliance but little commitment.

Activity: technical or adaptive?

Place each challenge into a bucket. Discuss in pairs for 3 minutes.

Technical	Adaptive	Mixed
Known fix; experts and a plan can solve it	People must change habits, beliefs or relationships	Technical solution plus adaptive barrier

1. "Replace diesel generators with a solar-battery system."
2. "Engineers and sales disagree about what customers value in repairable products."
3. "Cut data-centre energy use by tuning cooling."
4. "Suppliers resist sharing emissions data."
5. "Senior staff refuse to give up business-class travel."
6. "Switch the canteen to a reusable crockery system."

SECTION 2

Core Leadership Skills

A practical toolkit: seeing the long term, deciding under uncertainty, influencing others and acting ethically.

Vision and purpose: leading with meaning

- **Why it matters:** people follow a purpose they understand and believe in.
- **Link to culture:** a vision is shared only when co-created; leaders start the conversation, not finish it.
- **Be specific:** name the outcome, the date, the owner and what will be given up.
- **Stay consistent:** decisions under pressure show whether the purpose is real.
- **Say it simply:** if a mid-level manager cannot repeat it, it will not guide choices.

Strategic thinking across three time horizons

Horizon	Focus	Leader's question
Now (0-2 years)	Efficiency, compliance, quick wins	Are we doing the basics well, and showing results?
Next (2-5 years)	New products, processes and capabilities	What must we build now to be competitive then?
Beyond (5-15 years)	Business model and system change	What world are we preparing for, and what is our role?

Balance all three. A portfolio of only Horizon 1 projects is safe but strategically weak.

Worked example: an internal carbon price

PROBLEM Replace a boiler. Option A: gas boiler, capex Rs 50 lakh, opex Rs 10 lakh/yr, 800 tCO2/yr. Option B: electric heat pump, capex Rs 80 lakh, opex Rs 6 lakh/yr, 400 tCO2/yr. 10 years, 10% discount rate (illustrative).

1. Annuity factor (10 years, 10%) = 6.1446
2. No carbon price: PV(A) = 50 + 10 x 6.1446 = Rs 111.4 lakh; PV(B) = 80 + 6 x 6.1446 = Rs 116.9 lakh. A looks cheaper by Rs 5.4 lakh.
3. With Rs 2,000/tCO2: carbon cost A = 800 x 2,000 = Rs 16 lakh/yr; B = Rs 8 lakh/yr.
4. PV(A) = 111.4 + 16 x 6.1446 = Rs 209.8 lakh; PV(B) = 116.9 + 8 x 6.1446 = Rs 166.0 lakh. B is cheaper by Rs 43.7 lakh.
5. Break-even price = 5.42 lakh / (400 t x 6.1446) = about Rs 221 per tonne.
6. Leaders use the price as a decision tool, not as a tax: it makes a hidden cost visible.

Even a small carbon price flips this decision. The leader's job is to make the cost visible and to decide with it.

Influencing without authority



Build coalitions

Find allies in finance, operations and the front line before the big meeting.



Frame for interests

Speak in terms of each person's goals: cost, risk, pride, growth.



Use evidence

Numbers and a short story from a credible pilot.



Start with a pilot

Lower the risk; let results persuade sceptics.

Stakeholders, negotiation and trade-offs

- **Investors:** want returns and risk control; show the link to value.
- **Regulators:** want compliance and transparency; engage early.
- **Employees and unions:** want security and fairness in a transition.
- **Communities:** bear local impacts; consult and share benefits.
- **Negotiation approach:** identify interests behind positions, create options for mutual gain, and be clear about non-negotiables.

TRADE-OFFS ARE NORMAL

Cost vs. emissions, jobs vs. closure, speed vs. fairness. Make the trade-off explicit and explain the choice.

Communication and storytelling for leaders

- **Start with why:** explain the purpose before the plan.
- **Be honest:** share setbacks and uncertainty, not only wins.
- **Tell stories:** one concrete example moves people more than a slide of statistics.
- **Listen more than you speak:** ask what blocks people; act on what you hear.
- **Tailor:** adjust language for the board, the plant floor and the public.
- **Avoid:** doom, jargon and green claims you cannot back with evidence.

Ethical judgment: common dilemmas and questions to ask

Dilemma	Questions a leader asks
Cost vs. environmental performance	What is the full cost, including future risk? Who bears the harm if we choose cheaper?
Jobs vs. closing a polluting plant	Is there a just-transition plan? Have affected workers been consulted?
Marketing claims vs. evidence	Can we prove it? Would we be comfortable if a journalist checked?
Speed vs. consultation	Who is missing from the decision? What do we lose by rushing?

Resilience: leading through setbacks

- **Expect resistance and failure:** pilots fail, budgets get cut, allies leave.
- **Separate setback from purpose:** learn from the failure, keep the goal.
- **Protect energy:** sustainability work can feel endless and emotionally heavy; share the load, rest, and celebrate progress.
- **Build support:** peer networks, mentors and a coalition reduce burnout.
- **Model it:** teams copy how leaders respond to bad news.

Partnerships: transactional or transformative?

Transactional

- Donation or one-off sponsorship
- Brand benefit is the main goal
- Short projects, unclear impact
- Partners meet rarely

Transformative

- Shared goals and joint resources
- Both sides change how they work
- Multi-year, with learning built in
- Trust built through regular dialogue

Quick check 2

A leader convenes competitors, farmers and a local NGO to fix a shared water shortage that no single firm can solve. Which systems-leadership capability is most at work?

A	A. Seeing the larger system
B	B. Delegating tasks downward
C	C. Cost cutting
D	D. Compliance reporting

Think, vote, then discuss. (Answer in speaker notes.)

Quick check 2: answer

Answer: A. Seeing the larger system

No firm can solve a shared water problem alone. By bringing the actors together, the leader helps them see the whole system, reflect on each other's views and start to co-create options. This is also an example of working at the ecosystem level.

SECTION 3

Leading in Practice

Leaders in action: three cases, the roles leaders play and what this means for engineers.

Case: Patagonia and a founder's radical choice

VALUES-LED OWNERSHIP

- Sept 2022: founder Yvon Chouinard and family transferred ownership of Patagonia to a trust and a non-profit.
- Stated aim: profits not reinvested in the business go to fight the climate crisis and protect nature.
- The move came after decades of environmental activism and product-stewardship messages such as encouraging repair.
- Honest critique: the model works because of a founder with unique control; most listed firms cannot copy it, and the brand still sells new products.

LESSON FOR CHANGE LEADERS

A leader's values were encoded in ownership and governance, not only in slogans. That makes the commitment harder to reverse, but the approach is hard to replicate in public companies.

Case: Orsted's strategic transformation

LEADING A BUSINESS THROUGH A PIVOT

- Started as the Danish fossil-based energy company DONG Energy, with oil, gas and coal in the mix.
- Leadership made a deliberate shift: sold the oil and gas business, moved toward offshore wind and later renamed itself Orsted.
- Backed the strategy with long-term investment and a coal-exit plan.
- Honest critique: recent years brought cost pressures, project write-downs and supply-chain problems for offshore wind. Check the latest company reports.

LESSON FOR CHANGE LEADERS

Bold strategic leadership can reshape a company, but a transformation carries real financial and execution risk. Leaders must keep communicating honestly when results wobble.

Case: Mahindra and the internal carbon price

INDIAN CORPORATE LEADERSHIP

- 2016: Mahindra & Mahindra announced an internal carbon price, reported as the first Indian company to do so.
- The Group later committed to become carbon neutral by 2040.
- Reported milestones include a plant described as India's first carbon-neutral facility.
- Internal pricing turns climate risk into a number that managers can use in investment decisions.
- Verify the price level, scope and current status in Mahindra's own reports.

LESSON FOR CHANGE LEADERS

Leaders can build tools into capital allocation, so sustainability appears in ordinary decisions. Be precise about claims such as "carbon neutral": ask about scope and offsets.

Four roles a sustainability leader plays



Visionary

Paints a future worth working toward.



Convener

Brings the right people and rivals to one table.



Translator

Connects science, finance and operations languages.



Challenger

Questions comfortable assumptions and holds others to account.

Leading from where you are: engineers and technologists

- **Software and data:** measure the energy and carbon of your systems; design for efficiency and, where possible, carbon-aware scheduling.
- **Product decisions:** engineers shape repairability, materials and energy use long before a manager sees the design.
- **Speak up early:** raise sustainability trade-offs at design review, not after launch.
- **Build allies:** a small group with a prototype often beats a policy memo.
- **Be accurate:** honest numbers and clear assumptions are your credibility.

FOR YOU

Leadership is a habit built early: take one small, visible project and lead it end to end.

When sustainability collides with short-term profit

Weak responses

- Quietly drop the goal
- Blame the market
- Protect the quarter and promise "later"
- Hide the trade-off from staff

Strong responses

- Name the trade-off and the owner
- Reframe: cost of inaction, risk, option value
- Phase the commitment with clear gates
- Explain the decision openly to stakeholders

Governance: where leadership becomes accountable

- **Board oversight:** a named committee reviews climate and sustainability risk and performance.
- **Executive accountability:** targets in senior scorecards, with pay linked where it is meaningful.
- **Clear roles:** who owns each target, who can say no, who escalates.
- **Transparent reporting:** publish progress and misses using accepted standards.
- **Why it matters:** governance outlasts any one leader; it is what protects commitments after leadership turns over.

Worked example: a 360-degree assessment

PROBLEM A sustainability lead rates herself and is rated by colleagues on five competencies (1-5): Systems thinking 4.2 vs 3.4, Anticipatory 3.8 vs 3.6, Normative 4.5 vs 4.1, Strategic 3.9 vs 3.2, Interpersonal 4.0 vs 3.0 (self vs others).

1. Gaps (self - others): 0.8, 0.2, 0.4, 0.7, 1.0
2. Self mean = (4.2 + 3.8 + 4.5 + 3.9 + 4.0) / 5 = 20.4 / 5 = 4.08
3. Others' mean = (3.4 + 3.6 + 4.1 + 3.2 + 3.0) / 5 = 17.3 / 5 = 3.46
4. Average gap = 4.08 - 3.46 = 0.62 (check: 3.1 / 5 = 0.62)
5. Largest gap and lowest rating by others: Interpersonal (1.0, 3.0). Next: Systems thinking (0.8) and Strategic (0.7).
6. Action: invest in listening and stakeholder engagement first, with a mentor and a cross-functional project.

Self-ratings run higher than others' ratings. The biggest gap, not the highest score, tells you where to develop.

Discussion: leading a campus initiative

Scenario: you are a final-year student chairing a new campus sustainability council. You have no budget, little authority and sceptical wardens.

1. Is your challenge technical or adaptive? Why?
2. Which three stakeholders do you approach first, and with what message?
3. What quick win and what pilot would you choose?
4. Which of your own skills needs the most development?

SECTION 4

Developing Leaders

Leaders are made, not only born. Organisations and individuals can build the skills deliberately.

How leaders learn

- **Experience:** stretch assignments, leading real projects with real stakes.
- **Relationships:** mentors, peers and coaches; feedback from people who disagree.
- **Education:** courses and reading give frameworks, but only help when applied.
- **Reflection:** regular review of what worked, what failed and why.
- **Rule of thumb:** the popular "70-20-10" split (experience, others, courses) is a rough guide, not an evidence-based formula.

Development pathways

Method	What it builds	Example
Action-learning projects	Strategic and systems skills	Cross-functional team cuts site waste; reports to management
Job rotation	Understanding of silos and trade-offs	Operations engineer spends 6 months in procurement
Mentoring and coaching	Self-awareness, judgment	Senior leader mentors a young sustainability lead
Peer networks	Resilience and shared learning	Cross-company sustainability roundtable
360 feedback and reflection	Awareness of blind spots	Annual 360 with a development plan

Building a leadership pipeline

- **Identify early:** look for people who already convene, question and deliver across boundaries.
- **Include the frontline:** leaders emerge in plants, labs and offices, not only at headquarters.
- **Diversity:** varied backgrounds and perspectives improve problem-solving and legitimacy.
- **Reward the behaviour:** promotion criteria should recognise sustainability leadership.
- **Make time:** development fails when it is added to a full workload with no support.

Measuring leadership impact

Level	Indicators	Source
Behaviour	Frequency of leaders raising sustainability in decisions; 360 scores	Surveys, observation
Team outcomes	Participation, ideas implemented, trust and credibility scores	Pulse survey, platform data
Organisational results	Targets met, emissions, resource use, safety, retention	Operational data
Ecosystem	Supplier engagement, partnerships formed, policy contribution	Programme records

Leadership effects are hard to isolate. Triangulate several measures and avoid over-claiming.

Leadership derailers and remedies

Derailers

- The heroic leader who does everything
- Greenwash: words without action
- Burnout from over-commitment
- Ignoring those who disagree

Remedies

- Share ownership; build a team and successors
- Align claims with evidence and report misses
- Set limits, delegate, build support
- Seek dissent and test your assumptions

Critique: the limits of leadership-centred thinking

- **Heroic leader myth:** success depends on structures, incentives and collective action, not only on individuals.
- **Context constrains:** shareholder pressure, regulation and markets limit what a leader can do, as the BP and Unilever examples show.
- **Survivorship bias:** we study successful leaders and overlook those who tried the same and failed.
- **Power and fairness:** leadership talk can ignore who gains and who loses from a transition.
- **Evidence:** much of the sustainability leadership literature is conceptual or case-based; causal evidence is limited.

Your personal leadership plan

Horizon	Action	Evidence of progress
Next 30 days	Choose one real issue; map stakeholders; speak to three people who disagree	Stakeholder map; notes from conversations
Next 90 days	Run a small pilot with measurable results; ask for 360 feedback	Pilot data; feedback summary
Next 12 months	Lead a project end to end; find a mentor; build a peer network	Delivered result; named mentor
Beyond	Take on larger, cross-boundary challenges; develop others	Successors; wider partnerships

Key takeaways

1. Sustainability leadership is an activity anyone can exercise, not a job title.
2. Tell technical from adaptive challenges; most hard problems are adaptive.
3. Core skills: systems and long-term thinking, influence, negotiation, ethical judgment.
4. Tools such as an internal carbon price make hidden costs visible in decisions.
5. Cases show both what leaders can change and the limits they face.
6. Develop leaders through projects, feedback and reflection, and be critical of the heroic-leader story.

Collaboration and Partnerships

Working with stakeholders to achieve shared sustainability goals

Learning outcomes

- **Explain** why sustainability goals require collaboration, and the main partnership types.
- **Map** stakeholders using salience and the IAP2 engagement spectrum.
- **Design** a partnership: partner choice, governance, trust and collective impact.
- **Quantify** a shared-cost allocation, a stakeholder score and a partnership benefit-cost ratio.
- **Critique** cases and the limits of partnerships, including power and accountability.

SECTION 1

Why Collaborate

Climate, water, plastics and poverty are shared problems. No single firm, government or NGO can solve them alone.

Why no one acts alone

- **Shared problems:** emissions, water stress and waste cross company, sector and national boundaries.
- **Complementary strengths:** firms bring capital and scale, governments bring rules and legitimacy, NGOs and communities bring local knowledge.
- **Efficiency:** sharing infrastructure, data and learning lowers cost for each party.
- **Legitimacy:** solutions designed with those affected are more likely to be accepted.
- **Global framing:** SDG 17 calls for partnerships to deliver the other goals.

KEY IDEA

Collaboration is not charity or public relations. It is often the only way to reach goals that sit outside one organisation's control.

Shareholder view vs. stakeholder view

Shareholder primacy

- Firm exists to maximise owner returns
- Other groups matter only as means
- Short-term financial focus
- Collaboration seen as cost or risk

Stakeholder approach (Freeman, 1984)

- Firm creates value for many groups
- Those affected by or affecting the firm count
- Long-term, relationship-based
- Collaboration seen as a source of resilience

A spectrum of working together



Inform

Share information and updates.

Coordinate

Align schedules and avoid duplication.

Cooperate

Support each other's separate projects.

Collaborate

Plan and deliver jointly toward shared goals.

Integrate

Merge resources, governance or operations.

A teaching model. More integration is not always better: choose the level that fits the goal and the trust available.

Types of partnership in sustainability

Type	What it looks like	Main risk
Supplier collaboration	Joint emissions or safety programmes with suppliers	Power imbalance; cost shifted to small suppliers
Business and NGO	Standards, restoration or community projects	Reputation risk; greenwashing perception
Public-private	Infrastructure, waste, water or energy projects	Unclear risk sharing; slow procurement
Industry alliances	Shared targets, pacts and standards	Lowest common denominator; free riders
Community partnerships	Benefit sharing and local co-management	Tokenism; unequal voice
Academia-industry	Joint research, pilots and training	Different timelines and incentives

Benefits and costs of collaboration

Benefits

- Access to resources, skills and data
- Shared cost and risk
- Legitimacy and social licence
- Learning and faster scale

Costs and risks

- Time and coordination effort
- Loss of control and flexibility
- Free riders and unequal contributions
- Shared reputation damage if one partner fails

Worked example: sharing a treatment plant

PROBLEM Three firms need effluent treatment. Flows: A 200, B 300, C 500 m3/day. Standalone capex: A Rs 4 crore, B Rs 5 crore, C Rs 7 crore (total Rs 16 crore). A shared plant costs Rs 11 crore (illustrative).

1. Total saving = 16 - 11 = Rs 5 crore (31.25% of standalone cost).
2. Rule 1, split by flow (20%, 30%, 50%): A Rs 2.2, B Rs 3.3, C Rs 5.5 crore.
3. Savings: A 4 - 2.2 = 1.8 (45%); B 5 - 3.3 = 1.7 (34%); C 7 - 5.5 = 1.5 (21.4%). All are better off than going alone.
4. Rule 2, equal percentage saving: each pays 68.75% of standalone: A 2.75, B 3.44, C 4.81 crore (sum 11.0).
5. Which is fair? Rule 1 favours small users; Rule 2 shares gains equally. The choice is a negotiation, not a formula.
6. Stability check: no firm should be able to do better by leaving. Here, each pays less than standalone cost.

The partnership works only if every party gains. How the gain is shared is where collaboration is won or lost.

Quick check 1

A local fishing community has a legitimate claim and an urgent grievance against a port expansion, but little formal power. In Mitchell, Agle and Wood's salience model, this is a:

A	A. Dormant stakeholder
B	B. Dependent stakeholder
C	C. Definitive stakeholder
D	D. Discretionary stakeholder

Think, vote, then discuss. (Answer in speaker notes.)

Quick check 1: answer

Answer: B. Dependent stakeholder

Mitchell, Agle and Wood classify stakeholders by three attributes: power, legitimacy and urgency. Legitimacy and urgency without power is dependent: the group needs others (regulators, NGOs, the firm) to take up its claim. Good practice is to engage such groups directly rather than wait for them to gain power.

SECTION 2

Mapping and Engaging

Before you partner, know who is affected, who can affect you, and how deeply each should be involved.

Who are your stakeholders?

Internal

Employees, managers, owners, unions, students and staff.

Value chain

Suppliers, distributors, customers, contractors.

External

Regulators, investors, communities, NGOs, media, peers.

Silent

Nature and future generations: no voice, so someone must speak for them.

Stakeholder salience: power, legitimacy, urgency

Class	Attributes present	Engagement implication
Dormant	Power only	Monitor; may activate later
Discretionary	Legitimacy only	Keep informed; good for voluntary projects
Demanding	Urgency only	Listen; avoid over-reacting
Dominant	Power + legitimacy	Engage regularly; they expect influence
Dangerous	Power + urgency	Manage carefully; risk of coercion
Dependent	Legitimacy + urgency	Give voice; ensure their claim is heard
Definitive	All three	Top priority; engage immediately

Source: Mitchell, Agle and Wood (1997). Attributes change over time, so revisit the map regularly.

The IAP2 spectrum of public participation

Inform

Provide balanced information.

Consult

Gather feedback on options or decisions.

Involve

Work directly so concerns are understood and considered.

Collaborate

Partner in developing alternatives and choosing a solution.

Empower

Place final decision in the hands of the stakeholders.

Activity: choose the engagement level

Place each situation at the engagement level that best describes it. Discuss in pairs for 3 minutes.

Inform / Consult	Involve	Collaborate / Empower
One-way or limited feedback	Concerns shape the options	Joint or stakeholder decision

1. "A firm posts notice of a plant expansion with a comment form."
2. "Workshops with residents shape three design options; the firm decides."
3. "A village committee and the firm jointly choose the waste-site location."
4. "A company sends a newsletter about its new recycling scheme."
5. "A community trust decides how benefit-sharing funds are spent."
6. "A survey asks suppliers which support they would value."

Engagement methods

Surveys

Wide reach; limited depth.

Workshops

Dialogue and co-design in small groups.

Advisory panels

Standing groups for regular advice.

Site visits

See the issue first hand.

Grievance channels

A safe, trusted way to raise complaints.

Joint projects

Learn by working together.

Positions vs. interests

Positions (what they say)

- "No expansion near our village."
- "We cannot raise prices."
- "The regulator must act."
- Rigid; leads to win-lose bargaining

Interests (what they need)

- Safe water, livelihood, health, respect
- Margin, volume, predictable costs
- Fair rules, evidence, accountability
- Flexible; opens room for joint options

Worked example: scoring stakeholder salience

PROBLEM Score each group 1 to 5 on power (P), legitimacy (L) and urgency (U); salience = P + L + U (a simple screening score, illustrative).

1. Regulators: P 5, L 5, U 3, score = 13
2. Local community: P 2, L 5, U 5, score = 12
3. Investors: P 4, L 4, U 2, score = 10
4. Employees: P 3, L 4, U 3, score = 10
5. Suppliers: P 3, L 3, U 2, score = 8
6. Insight: the community ranks second despite low power. A power-only view would put it last and miss the biggest social licence risk.

Use scores to start a conversation, not to end one. Always check with the groups themselves.

Communities and free, prior and informed consent

- **FPIC:** free, prior and informed consent is a principle in the UN Declaration on the Rights of Indigenous Peoples and in many corporate and finance standards.
- **Free:** no coercion or manipulation.
- **Prior:** before decisions and in good time.
- **Informed:** full, understandable information, including risks.
- **Consent:** the community may say no. Consultation without the option to refuse is not consent.

WHY IT MATTERS

Projects that ignore local rights face delays, litigation and loss of social licence, and may cause real harm. Treat it as a standard of respect, not paperwork.

Case: Kalundborg Symbiosis

INDUSTRIAL COLLABORATION OVER DECADES

- Denmark: a cluster of companies and the municipality exchange waste heat, water, gypsum, steam and by-products.
- Exchanges grew step by step from the 1960s and 1970s, mostly on bilateral commercial deals.
- Each link is justified by cost savings and resource use; the environmental gain follows.
- Honest critique: it grew from specific local conditions and geography, so it is hard to copy elsewhere.
- Verify current members and quantities on the Symbiosis website.

LESSON FOR CHANGE LEADERS

Trust built through small commercial exchanges can grow into a system. Start with deals that make economic sense for both parties; do not wait for a grand plan.

Quick check 2

A company forms a joint committee with community representatives to develop alternatives and choose the preferred solution together. On the IAP2 spectrum this is mainly:

A	A. Inform
B	B. Consult
C	C. Collaborate
D	D. Empower

Think, vote, then discuss. (Answer in speaker notes.)

Quick check 2: answer

Answer: C. Collaborate

IAP2 defines Collaborate as partnering with the public in each aspect of the decision, including developing alternatives and identifying the preferred solution. Empower goes further by placing the final decision in the stakeholders' hands.

SECTION 3

Building Partnerships

How to find the right partners, set up good governance, build trust and deliver collective impact.

The partnership lifecycle



Scope

Agree the problem, goals and who needs to be at the table.

Build

Select partners, define roles, governance and resources.

Manage

Deliver, communicate, resolve conflicts, measure.

Renew or exit

Review results; expand, adapt or end responsibly.

Plan the exit at the start. A partnership with no end or review point tends to drift.

Choosing partners: what to look for

Criterion	Questions to ask	Red flag
Shared goal	Do we want the same outcome, even for different reasons?	Hidden agenda; partner wants only publicity
Complementary assets	What does each side bring that the other lacks?	Duplicated strengths; unclear contribution
Credibility	Track record, transparency, reputation?	History of broken promises
Capacity	Staff, time and money to follow through?	Over-committed; one champion only
Values and ethics	Compatible standards on rights, anti-corruption, environment?	Conflicts on core principles

Collective impact: five conditions (Kania and Kramer)

Common agenda

Shared vision and understanding of the problem.

Shared measurement

Agreed indicators and data across partners.

Mutually reinforcing

Different activities that coordinate into one plan.

Continuous communication

Regular contact to build trust and alignment.

Backbone support

A dedicated team to coordinate the effort.

Governance and agreements

- **Charter or MoU:** purpose, scope, roles, duration and how decisions are made.
- **Decision rules:** who votes, who vetoes, how deadlock is resolved.
- **Resources:** who contributes money, staff and data; how costs and benefits are shared.
- **Data and IP:** what can be shared, who owns results, confidentiality.
- **Disputes and exit:** a clear path for conflict and a fair way to leave.
- **Accountability:** regular reporting to partners and, where relevant, the public.

Building trust between partners



Transparency

Share goals, constraints and results, including bad news.



Reliability

Do what you promised, on time.



Reciprocity

Both sides give and take; no one only extracts.



Early joint wins

Small successes prove the partnership can deliver.

Power imbalance in partnerships

Dominated partnership

- The large partner sets goals and terms
- Small partners bear compliance costs
- Voice is token; exit costs are high
- Result: resentment, hidden non-compliance

Equitable partnership

- Goals set jointly; small partners heard
- Costs and support shared fairly
- Real decision rights and fair terms
- Result: durable commitment

Working with suppliers: audit or develop?

Audit and punish

- Checklists and inspections
- Sanctions for non-compliance
- Suppliers hide problems
- Compliance at best

Develop and partner

- Training, finance and technical help
- Joint improvement targets
- Honest reporting of problems
- Capability and loyalty grow

Business and NGO partnerships: benefits and risks

Side	What it gains	What it risks
Business	Credibility, expertise, access to communities, new ideas	Accusations of greenwashing; public criticism by NGO
NGO	Funding, scale, access to supply chains and data	Loss of independence; reputation by association
Both	A tested solution that neither could build alone	Mismatch in timelines, language and expectations

Mitigate with clear conflict-of-interest rules and the right for the NGO to speak publicly.

Case: India Plastics Pact

INDUSTRY ALLIANCE WITH SHARED TARGETS

- Launched in 2021 by CII and WWF India, with support from WRAP, as part of a global Plastics Pact network.
- Brings together businesses, government, NGOs and others around shared packaging goals.
- Typical elements: eliminating unnecessary plastic, making packaging reusable or recyclable, using recycled content, and reporting progress.
- Honest critique: voluntary pacts have limited enforcement, and progress depends on member reporting and on collection and recycling infrastructure.
- Check the Pact's latest targets and progress reports.

LESSON FOR CHANGE LEADERS

Shared targets and common reporting create peer pressure. They do not replace regulation or the waste-management system that the targets rely on.

Case: RSPO and the limits of multi-stakeholder standards

HONEST CRITIQUE

- The Roundtable on Sustainable Palm Oil began in 2004 with producers, retailers, banks and NGOs such as WWF.
- It created certification standards for more responsible palm oil.
- Strength: it brought rivals and critics to one table and set common criteria.
- Criticism: some NGOs and researchers argue that certification has not stopped deforestation or labour abuses in all cases, and that enforcement and audits vary.
- Check recent reviews and the RSPO's own reports for current evidence.

LESSON FOR CHANGE LEADERS

A seat at the table does not guarantee impact. The design of standards, auditing and consequences decides whether collaboration changes behaviour on the ground.

Worked example: benefit-cost of a partnership

PROBLEM A clean-cooking programme: business Rs 2 crore, government Rs 3 crore, NGO in-kind Rs 0.5 crore. It reaches 40,000 households, each saving Rs 3,000/yr in fuel for 5 years; discount rate 10% (illustrative).

1. Total cost = 2 + 3 + 0.5 = Rs 5.5 crore; cost per household = 5.5 crore / 40,000 = Rs 13,750
2. Annual saving = 40,000 x 3,000 = Rs 12 crore
3. Annuity factor (5 years, 10%) = 3.7908, so PV of savings = 12 x 3.7908 = Rs 45.5 crore
4. Benefit-cost ratio = 45.5 / 5.5 = 8.3; NPV = 45.5 - 5.5 = Rs 40.0 crore
5. Business share of cost = 2 / 5.5 = 36%; leverage = 5.5 / 2 = 2.75 times the business money
6. Caution: claiming 36% of the impact for 36% of the cost is crude. Report shared contribution, not exclusive credit.

Each partner's money was leveraged. The programme works only because all three contributed.

Discussion: partnerships for the IIT Delhi campus

Scenario: the campus wants to cut waste going to landfill by half. Waste is collected by a municipal contractor, hostels by wardens, and food by outsourced caterers.

1. Map the stakeholders. Who is dependent, dominant or dormant?
2. Which partners (municipality, NGO, recycler, students) would you invite first?
3. What shared target and measure would you agree on?
4. How would you handle the contractor, who has little incentive to change?

SECTION 4

Making It Work

Partnerships fail more often from poor management than from bad intentions. This section is about keeping them effective and honest.

Shared goals and shared measurement

- **Specific, joint targets:** what, how much, by when, owned by whom.
- **Common definitions:** agree on boundaries, units and methods before collecting data.
- **Avoid double counting:** when several partners claim the same result, the total is overstated.
- **Data-sharing agreements:** who sees what, how it is protected.
- **Report openly:** successes, misses and lessons, on a regular schedule.

Communication and conflict resolution

- **Regular rhythm:** agreed meetings and a single place for decisions and records.
- **Common language:** explain jargon; firms, NGOs and communities use words differently.
- **Interest-based negotiation:** separate people from the problem; focus on interests; generate options.
- **Escalation path:** agree in advance who mediates and when.
- **Listen to dissent:** objections often expose real risks.

Why partnerships fail, and safeguards

Common failure modes

- Unclear goals or roles
- Dominant partner and silent others
- No dedicated coordination capacity
- Loss of the founding champion

Safeguards

- Written charter, reviewed regularly
- Fair decision rights and real voice
- A funded backbone or secretariat
- Shared ownership and succession plan

Measuring partnership health

Dimension	Indicator	Method
Trust	"I trust partners to keep commitments" (1-5)	Anonymous partner survey
Participation	Attendance, contributions delivered on time	Records
Equity of voice	"My views influence decisions" (1-5), by partner type	Survey, by group
Outcomes	Progress on shared targets; cost per outcome	Shared dashboard
Learning	Changes made because of lessons	Review notes

Law and ethics: competition and greenwashing

- **Competition law:** rivals who coordinate on prices, output or sensitive commercial data may breach competition rules. Sustainability goals do not automatically exempt such agreements.
- **Safe practice:** keep sustainability collaboration pre-competitive and focused on standards, shared infrastructure or public goals; take legal advice.
- **Greenwashing risk:** joint claims must be accurate and evidenced; every partner is exposed.
- **Anti-corruption:** screen partners, especially in public-private deals.

CHECK BEFORE YOU SIGN

Rules differ by country and change over time. Treat this slide as a prompt to ask a lawyer, not as advice.

Critique: the limits of partnerships

- **Accountability gap:** when many partners share responsibility, no one may be answerable for failure.
- **Power asymmetry:** big firms and funders often set the agenda; communities and small suppliers have less voice.
- **Partnership as PR:** a logo on a partnership page can substitute for real change.
- **Substitute for regulation:** voluntary collaboration may delay binding rules that would level the field.
- **Evidence:** much of the evidence is case-based and written by participants; independent evaluation is rarer.